General instruction: For each slide listed here, you should bound its content within that slide only

Slide 1: About Project SELARAS
Note: Use arrows to show the situation -> problem statements -> impact as a progression

Situation	Problem Statements	Impact
BNM's credibility as a regulator rests on being **internally consistent** with its own body of policy and **externally aligned** with international standards	Manual assessment of overlapping areas of concern with other departments	Automatic detection of overlaps with other departments
	Manual benchmarking of BCBS and other jurisdictions' policies	Optimised opportunity-gap analysis to explore and exploit established practices
	Lengthy drafting lifecycle where policy drafters spend disproportionate time reading and executing routine tasks instead of higher-value work	Drafters' time reallocated to higher-value activities, e.g. review and scoping what needs to be assessed
	Institutional awareness and domain expertise take years to develop, and are concentrated in individual drafters	Institutional awareness and domain expertise decoupled from individual drafters and embedded into the system

Slide 2: Specialised agent …
Note: Backend engine extraction -> chunking -> retrieval (what, how) combo of system design + business rules

Slide 3: Feasibility and Scalability

	Feasibility	Scalability
Functional	• **UX is accurate to the user journey of policymaking** Feature: Node and edge analysis support the	• **Extending use case to other departments or domains does not require further configuration**

	lifecycle of discussion decks -> discussion paper -> exposure draft -> policy document ● **Policy recommendations are practical** Evidence: 67% of sampled recommendations are marked as valid by an FDI analyst Feature: Recommendations are classified based on input policy requirement ● **Citation integrity is guaranteed** Feature: Finder pipeline is forced to return findings with citations, unsupported findings are strictly dropped	**Evidence:** Onboarding new departments only requires document preparation and ingestion rather than additional application logic
Non-functional	● **LLM cost is optimised** **Evidence:** Small model (Haiku) for concept retrieval, Medium model (Sonnet) for batched same-topic finder (differs-on, conflicts-with, aligns-with), Large model (Opus) for whole document coverage finder (silent-on, goes-beyond)	● **Cost scales with edges actually analyzed, not with graph size** Feature: Linkage analysis is dynamically evaluated on-demand by user, so growth is linear and bounded by demand, preventing exponential computational overhead.

Slide 4: Risks and Mitigation

Risks	**Mitigation**
The AI gets it wrong - A finding cites a clause that doesn't exist, or misjudges one that does	Citations are passed into a deterministic validation guard, not an AI model. The drafter can also accept or reject any finding.
Confidentiality - Bank drafts or internal papers leave the Bank	[In scope] Prototype built using public documents for demonstration. [Out of scope] Hosting on the Bank's private infrastructure, deploying private AI models, and building an enterprise-grade application.
Accountability - A policy ships with an error and nobody owns the decision	[Phase 2] Maker-checker with a full audit trail. Enforce segregation of duties where review actions need a checker and a checker cannot be the reviewer .

Slide 5: Implementation Timeline
Note: Present this slide with arrow progression.

Phase 1 - Prototype (Delivered)	Phase 2 - Next Steps
• **AI linkage-finding engine, live on the Open Finance policy workstream** Proves the model correctly cross-references a real BNM policy document (Exposure Draft) against external reference standards (RMiT, BIS, HKMA, ABS/MAS API playbook). • **Five-label semantic classification with verbatim clause citation (aligns-with / differs-on / conflicts-with / silent-on / goes-beyond)** Every finding quotes its source clause; zero fabricated citations by design. • **Knowledge graph of the workstream (documents as nodes, structural edges)** Replaces manual cross-referencing of policy documents with a navigable map. • **AI drafting Copilot** 5-step workflow (/explore → /brainstorm → /draft → /write → /deliver) writes citation-grounded recommendations straight into the draft.	• **Deduplicated institutional graph** Scales the one-workstream prototype into a BNM-wide regulatory map; merge all workstreams into one network where shared entities collapse to a single node. • **Graph database + agentic Cypher retrieval** Natural-language questions traverse the whole institution's graph in one query, can answer questions like *“Which of our outstanding policy documents are silent-on alternative collateral (movable assets, cash flow-based, invoice financing) that peer regulators (MAS, HKMA) have already codified?”* → pinpoints exactly where our rulebook never gave banks permission to lend against what SMEs actually have” • **Jurisdiction / BIS clustering view** Group the graph by regulator and Basel standard; converts an internal linkage tool into a benchmarking answer to "how does BNM compare globally." • **Maker-checker with a real per-finding audit trail** Enforced maker≠checker, every transition logged with actor/timestamp/comment. • **M365 / SharePoint integration** Pull PDF/PPTX/XLSX directly from existing SharePoint sites, alternative to manual upload.

Slide 6: Reusability of Project SELARAS

<table>
<tr><th>Depart ment</th><th>Opportunities</th><th>Workstream / BAU</th></tr>
<tr><td>FDI</td><td rowspan="4">International Regulatory Convergence:
Automated benchmarking of BNM’s policies against BIS, MAS, and HKMA standards. (node_type: international-standards)

Cross-Departmental Overlap:
Surfacing overlapping or conflicting requirements across departments prior to publication (node_type: internal-published)

Alignment to Act / Law:
Verify that playbooks and blueprint comply with governing legislation (node_type: act-law)</td><td>● Open Finance PD</td></tr>
<tr><td>FS</td><td>● Banking System Financial Crisis Management Framework (Crisis Playbook)
● Conforming Loans Standards (CLS)
● Financial sector blueprint (FSBP) for 2027-2030</td></tr>
<tr><td>PFP</td><td>● Operational Resilience Roadmap</td></tr>
<tr><td>PPD</td><td>● DP on stablecoins and CBDC
● CEO letter on tokenised deposit</td></tr>
<tr><td>JP4</td><td>Supervisory Coherence:
Checks new policy drafts against past supervisory letters issued on RH platform to ensure consistent guidance to financial institutions. (node_type: supervisory-letter)</td><td>● Capital Adequacy Framework (Internal Ratings-Based Approach for Credit Risk) PD
● Capital Adequacy Framework
● Interest Rate Risk in the Banking Book PD</td></tr>
<tr><td>JDM</td><td>MPComm Coherence:
Benchmarks new drafts against past statements to maintain consistent messaging around OPR</td><td>● Economic Monetary Review (EMR)
● Monetary Policy Statement (MPS)</td></tr>
</table>

	decisions (hike, maintain, cut).	

Slide 7 User Feedback:
User Feedback
Note:
- **Call out statement: 67% of sampled recommendations were assessed to be valid by a FDI Policy Analyst.**
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Recommendation	**Rationale**	**Action for BNM**	**Comments**
Add a public TSP registry and anti-impersonation requirement	HKMA requires banks to publish all partnering TSPs and their products, and to monitor for fraudulent parties impersonating TSPs. ED requires TSP oversight but has no provision on publishing that list or on impersonation monitoring.	Add a requirement for data providers to maintain and publish an up-to-date list of authorised TSPs accessing their systems, and an obligation to warn customers against impersonation fraud.	****The gap is relevant****, as a public registry of data consumers is useful, and Malaysia does not currently have one. However, it may be less relevant in Malaysia's context as participation is currently limited to FIs and does not yet include fintech-type participants. Also, I checked the Hong Kong framework. Although HKMA uses the term "TSP" (third party service provider) and our ED uses the same term, they refer to different things. HKMA's TSP is closer to a data consumer in Malaysia, as it refers to an

			entity that accesses banking data to provide services. In contrast, Malaysia's TPSP refers to a third party engaged by an FI to support its operations (e.g. cloud, data analytics)
Consider extending data-sharing obligations to non-bank and big-tech players	BIS identifies that banks must share customer data under open finance mandates while non-banks, fintechs, and big-techs holding financial-sector-relevant data face no equivalent obligation — creating market distortions. ED applies exclusively to defined FSPs.	Assess whether a subsequent phase should extend reciprocal data-sharing obligations to non-regulated data holders, or explicitly state that the current asymmetry is an accepted design choice.	****The recommendation is good****, but ED currently only covers banking participants and does not include non-bank participants, the issue of unequal data-sharing obligations is not applicable

Maintain and promote ED's consent infrastructure as a regional benchmark	ED materially exceeds both HKMA and BIS on consent protection: 7-year audit logging with customer-accessible logs, a mandatory real-time consent dashboard, and automatic data deletion on consent expiry — none of which peers require.	Position these obligations explicitly as consumer-protection features in public communications. Resist any future consultation pressure to streamline consent requirements. Consider publishing a regional comparison to evidence Malaysia's leadership.	**This is valid strength**
Clarify which RMiT technology requirements apply to open finance participants	RMiT requires comprehensive cryptography policy (encryption standards, key lifecycle, compromise-recovery) and an enterprise technology architecture framework with SDLC methodology. ED cross-references RMiT generally but does not map which provisions are activated by open finance participation, creating compliance ambiguity.	Publish a mapping annex or guidance note specifying the RMiT sections directly applicable to open finance data providers and consumers, particularly for smaller FSPs that may not have previously implemented full RMiT coverage.	Usually BNM PD likes to reference other PD (e.g. RMIT) in general terms rather than citing specific provisions, because the provision numbering may change when those documents are updated. To avoid the OF ED becoming outdated, it is intentional that we do not map requirements to specific RMiT provisions.

Strengthen the phased implementation timeline with concrete milestones	ED describes a gradual phased approach but is vaguer than HKMA (concrete phases with prescribed timelines) and more demanding than BIS jurisdictions (voluntary/facilitative roadmaps). Malaysia's mandatory approach is a deliberate policy tighten but places the burden on BNM to specify expectations per phase.	Adopt HKMA's model of prescribing concrete deliverables and timelines per phase. Reduce FSP uncertainty by publishing a phase transition checklist and specifying what constitutes compliance at each gate.	It is true that HKMA has more details on their milestone and **it is a valid recommendation** too (but FYI we intentionally did not prescribe the milestones further in the ED, as PayNet, in its role as the platform operator, is already working through the detailed implementation plan with the pilot banks through existing industry engagements -- and for future, participants can go to them on implementation details. BNM PD is meant to be principle-based if possible)
Reconcile the open finance breach-response regime with RMiT's incident-response regime	ED's escalation focuses on containing and remediating customer information breaches. RMiT's escalation focuses on service resumption and performance stabilisation. An open finance breach will simultaneously trigger both, but the interaction is unspecified.	Add a cross-reference or guidance note clarifying how the two escalation chains interact — which takes precedence, who leads, and how FSPs report to BNM when a single incident activates obligations under both instruments.	We referenced MCIPD (another policy document) on requirements for customer information breach reporting. I don't know how relevant is this for open finance and I also have not considered whether the requirements under MCIPD and RMiT should be aligned, cause these two PDs are owned by different departments (CMC and TCS). That said, **this could be a**

			valid recommendation for further consideration.
Monitor voluntary-jurisdiction experience to calibrate phase sequencing	Malaysia's mandatory, fixed-date rollout diverges from the facilitative model BIS observes in most jurisdictions. BIS's international evidence shows jurisdictions that mandated too fast — without market readiness — faced low TSP participation and poor API quality.	Build formal readiness checkpoints into each phase transition using FSP readiness assessments and TSP onboarding metrics before activating the next phase obligation.	Valid recommendation

Consider converting the FSP financial literacy encouragement into a requirement	ED encourages FSPs to develop simplified digital and financial literacy guides on open finance and consent management. HKMA relies only on regulator-led education. Given the sophistication of ED's consent framework relative to average consumer awareness, voluntary guides are likely insufficient.	Elevate the guidance language from encouraged to a supervisory expectation or minimum standard, particularly for FSPs targeting retail and underserved segments.	Valid recommendation (FYI we have communication workstream with pilot group banks and hence it's not further specified in ED, but yea valid concerns) But I don't know if this recommendation can be considered as Strength, seems more like a Gap

Foundational design validated — no changes required on core alignment areas	Multiple findings confirm ED aligns with both HKMA and BIS on liability frameworks, data protection compliance obligations, technical safeguards for customer information, and API-based data sharing architecture.	No structural changes indicated. Use these alignment findings as evidence in stakeholder consultations where FSPs may challenge the compliance burden — the framework's foundations are consistent with international best practice.	While the statement is fairly accurate, BIS does not actually provide much detail on e.g. liability arrangements, so there is not much to compare against. Our ED is also relatively high-level on liability, which is why it appears to be a match. It may be helpful to state this explicitly, otherwise readers could get the impression that our approach is strongly aligned with BIS standards when the comparison is largely due to the absence of detailed requirements on both sides.

Slide 8:

Cost Estimation

Cost Per Document-Pair Analysis *(Assumptions below, not measured)*

Stage	Model	Assumed Volume	Assumed Tokens	Cost
Stage 3: Batched	Sonnet 5 ($3/$15 per MTok)	8 calls (60 candidates $\div$ 8/call)	~3,000 in / 800 out per call	$0.168

same-topic finder				
Stage 5: Whole-doc coverage finder	Opus 4.8 ($5/$25 per MTok)	1 call	~8,000 in / 1,500 out	$0.078
Total per document pair				~$0.25
Stage 1: Axis extraction *(One-time per document, cached)*	Haiku 4.5 ($1/$5 per MTok)	1 call per document	~2,000 in / 200 out	$0.003/doc

- Deterministic Pipeline Stages: Stages 2, 4, and 6 are deterministic code (retrieval, suppression, merge/citation-validation) with no model cost.
- Corpus Generation: Generating the entire committed demo corpus (the 9 recorded document pairs plus one-time extraction across 9 documents) costs under $3 total, which is exactly why the demo strategy is build-and-persist rather than live-on-stage.

- Copilot Q&A: A copilot Q&A turn (Sonnet, ~4,000 in / 500 out) runs about $0.02: cheap enough that interactive use during judging is a non-issue.
- MVP Cost Today: Effectively negligible: low single-digit dollars in model spend for the whole committed fixture set, plus whatever the FastAPI + Vite hosting costs during the hackathon window (a free-tier or single small instance covers it).

Production Estimate (AWS Hosting)

Model Inference

Scaled to the ~200–240 policy staff across the five to six departments named in the Reusability analysis (assuming 5 new analyses + 20 copilot turns per user per week: a planning assumption, not a measured rate)

Item	Volume / Month	Cost / Month
Document-pair analyses	~4,330 (200 users $\times$ 5/week $\times$ 4.33)	~$1,080
Copilot turns	~17,300 (200 users $\times$ 20/week $\times$ 4.33)	~$345
Embeddings (retrieval index)	Re-indexing new/changed documents	~$50

Model Inference Subtotal		~$1,500 / month

Where Those Model Calls Run on AWS Matters

Two different things share the "AWS" label:

- Claude Platform on AWS: Anthropic-operated, same-day feature parity with the first-party API, same pricing table above, AWS-native billing/IAM. This is the closer match to what's already built and tested against first-party pricing.
- Amazon Bedrock: AWS-operated, its own release cadence and a separate Bedrock pricing table, `anthropic.`-prefixed model IDs. Don't assume 1:1 pricing parity with the numbers above: check Bedrock's own pricing page before committing to it.

Recommendation: Given the codebase already targets first-party Claude behavior (`engine/config.py` deployment names, no Bedrock-specific handling), Claude Platform on AWS is the lower-migration-risk option if AWS hosting is a hard requirement.

Non-Model AWS Infrastructure

Sized for this traffic level (a few hundred users, not high-throughput)

Component	Purpose	Cost / Month
ECS Fargate *(or Lambda, given spiky usage)*	FastAPI backend	~$50–70
S3 + CloudFront	Vite/React frontend, static hosting	~$10–15

S3	Corpus, clause index, findings JSON	~$5
RDS (db.t3.micro) *(or Aurora Serverless v2)*	If multi-user review state needs a real DB (currently fixture JSON)	~$50–100
ALB + CloudWatch	Load balancing, logs/metrics	~$30
Infra Subtotal		~$150–220 / month

Bottom Line

Combined Production Estimate: Roughly $1,650–1,800/month to run SELARAS for the entire ~200-person target user base across five-plus departments: against a status quo of weeks of manual analyst time per policy document. The Message Batches API (50% off, for non-interactive work like overnight re-indexing) is an easy further lever if usage grows past this estimate.

Slide 9: Limitations (for end to end production)

- **LLM timeout when document is too large, limited context window**
- **Non-pdf file (e.g. xlsx, pptx) chunking performance may be poor, noisy and inconsistent**
- **Dependency on parsable document formats (scanned with poor PDF quality, complex column layout can degrade parsing quality)**
- **Coverage findings (i.e. silent-on, goes-beyond) may be noisy on documents that are related but topically differ**

Slide 10: Limitations of this prototype

- **Copilot and rulebook are currently running with pre-filled data**
- **Currently only support PDF, not excel, PPTX**
- **Chunking (implemented structured-rules, unstructured, prose exposed on frontend selection no routing logic)**